AMHARIC ACOS QUADRUPLE EXTRACTION

Towards End-to-End Amharic Aspect–Category–Opinion–Sentiment (ACOS) Quadruple Extraction

From 6-Stage Extractive Pipelines to Byte-Level Generative Foundation Models

Author Azariyas Mekonen

Target Domain Amharic Civic & Governance Commentary

Current State SDRN: Synchronous Double-Channel Recurrent Network

Source github.com/EthAIResearch/acos-amharic

PHASE 2 DEEP-DIVE | Theory, Mathematics & Algorithmic Mechanics

Joint Aspect-Opinion Pair Extraction (AOPE) in Amharic Civic Commentary

October 2026 · Technical Report

The Aspect-Opinion Pair Extraction (AOPE) Task

Formulation of the joint pair extraction problem and why pipelining fails.

Task Definition & Objectives

Input: Amharic sentence composed of N lexical words or tokens.

Target Output: Set of explicit aspect-opinion pairs (Aspect, Opinion).

Aspect span: Target governance entity, service, or policy mentioned in text.

Opinion span: Evaluative predicate or sentiment expression targeting the aspect.

Example: [የከተማው አስተዳደር (Aspect) | ጥሩ ስራ ሰርቷል (Opinion)].

Coupled semantics: An opinion is only meaningful relative to its aspect target.

Why Pipelining Fails (The 14% Collapse)

Stage 1 (Span Extraction): Detects aspects and opinions in isolated passes.

Stage 2 (Heuristic Pairing): Applies nearest-token word distance rules.

Failure: Nearest-distance achieves 99.3% F1 ONLY on perfect gold spans.

Against noisy Stage 1 spans, Stage 2 has zero recovery mechanism.

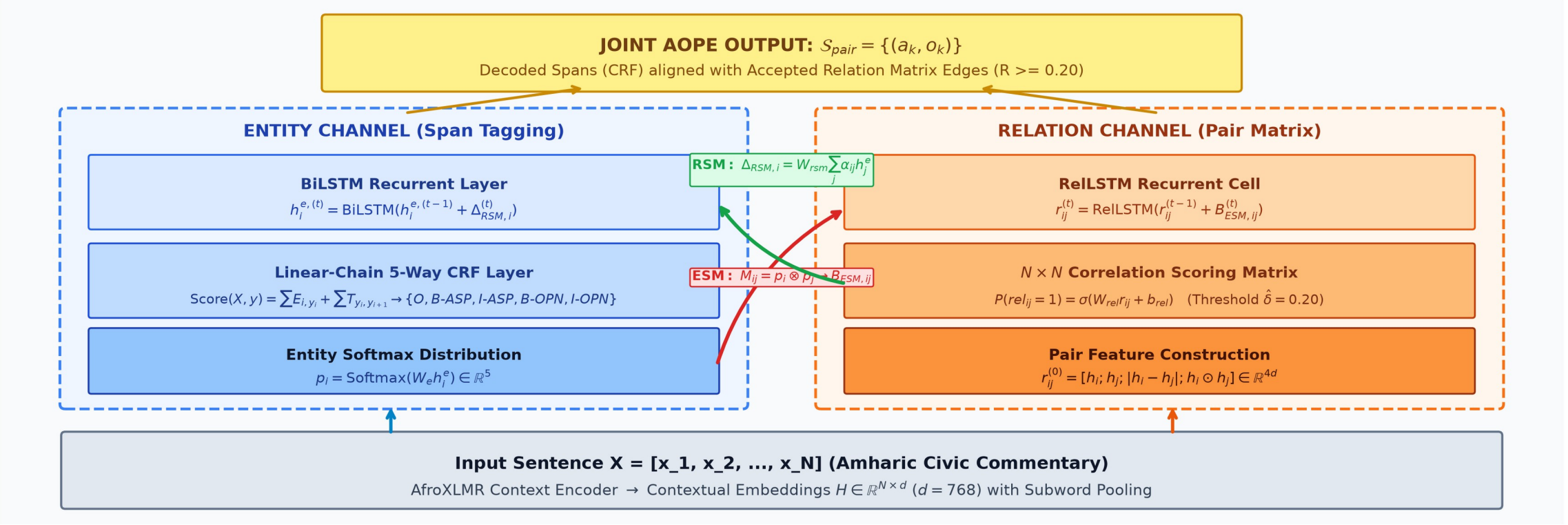
Upstream false negatives and boundary shifts cascade irreversibly.

End-to-end full quadruple F1 collapses to approximately 14.2%.

AOPE requires joint modeling: an opinion cannot be reliably extracted without simultaneously identifying its aspect target.

Dual-Channel Recurrent Interaction with ESM & RSM Loops

Complete architecture diagram illustrating synchronous feedback between Entity and Relation channels.



Non-synchronized models fail; only bidirectional synchronization between entity and relation channels outperforms pipelines.

Contextual Sequence Representation & Subword Pooling

Mathematical formulation of the contextual text encoder layer.

AfroXLMR Contextual Encoding & Subword Pooling Formulation

$$H = \text{AfroXLMR}([x_1, \ldots, x_N]) \in \mathbb{R}^{N \times d}, \quad \text{where } d = 768$$

$$H_{word} = M_{align} H_{sub}, \quad M_{align} \in \mathbb{R}^{N_{word} \times N_{sub}} \quad \text{(Word-Level Masked Mean)}$$

- Why AfroXLMR: Pretrained on multilingual African corpora, capturing long-distance Amharic SOV dependencies.
- Why Subword Pooling: Averages constituent wordpiece vectors to align representations with authentic lexical words.

Contextual encoding establishes continuous 768-dimensional semantic vectors for both downstream recurrent channels.

5-Way BIO Sequence Tagging Formulation

Mathematical formulation of the Entity Channel's hidden states and emission scores.

SDRN Entity Channel: Recurrence, Emissions & 5-Way Tag Space

$$H_e^{(0)} = H \in \mathbb{R}^{N \times d}, \quad h_i^{e,(t)} = \mathrm{BiLSTM}\left(h_i^{e,(t-1)} + \Delta_{RSM,i}^{(t)}\right)$$

$$E_i^{(t)} = W_e h_i^{e,(t)} + b_e \in \mathbb{R}^5, \quad \mathcal{Y} = \{O, B\text{-}ASP, I\text{-}ASP, B\text{-}OPN, I\text{-}OPN\}$$

- Why 5-Way BIO Tagging: Unifies aspect and opinion extraction into mutually-exclusive labels, preventing dual-tag conflicts.
- Why BiLSTM Recurrence: Models bidirectional word sequence context while dynamically absorbing relational feedback Δ_{RSM}.

The entity channel models sequential boundary tags while continuously accepting relational feedback from the relation channel.

5. RELATION CHANNEL MATH

N × N Token-Pair Correlation Matrix Formulation

Mathematical formulation of pairwise token affinity representations.

SDRN Relation Channel: $N \times N$ Token-Pair Correlation & RelLSTM

$$r_{ij}^{(0)} = [h_i\,;\,h_j\,;\,|h_i - h_j|\,;\,h_i \odot h_j] \in \mathbb{R}^{4d}, \quad S_{ij} = \frac{h_i^T W_r h_j}{\sqrt{d}}$$

$$r_{ij}^{(t)} = \mathrm{RelLSTM}\left(r_{ij}^{(t-1)} + B_{ESM,\,ij}^{(t)}\right), \quad P(rel_{ij} = 1) = \sigma\left(W_{rel} r_{ij}^{(t)} + b_{rel}\right)$$

- Why $N \times N$ Matrix: Directly evaluates all pairwise word relations, bypassing surface distance in Amharic's SOV structure.
- Difference & Product Terms: $|h_i - h_j|$ captures feature contrast; $h_i \odot h_j$ models multiplicative affinity.

The relation channel explicitly scores whether token pairs are semantically connected, bypassing linear word distance.

Entity-to-Relation (ESM) & Relation-to-Entity (RSM) Feedback

Mathematical formulas governing bidirectional cross-channel feedback loops.

SDRN Synchronous Bidirectional Feedback Equations (Chen et al. 2020)

$$\textbf{ESM:} \quad p_i = \mathrm{Softmax}(W_e h_i^e) \in \mathbb{R}^5, \quad M_{ij} = p_i \otimes p_j, \quad B_{ESM,ij} = W_{esm}\mathrm{vec}(M_{ij})$$

$$\textbf{RSM:} \quad \alpha_{ij} = \mathrm{Softmax}_j(W_\alpha r_{ij}), \quad c_i = \sum_{j=1}^{N} \alpha_{ij} h_j^e, \quad \Delta_{RSM,i} = W_{rsm} c_i$$

- Why ESM is used: Suppresses relations involving background *O* tokens, forcing the relation matrix to focus on entities.
- Why RSM is used: Pulls target aspect semantics into ambiguous opinion verbs, rescuing dropped boundaries.

ESM filters spurious non-entity relations, while RSM pulls aspect evidence into ambiguous opinion predicates.

The Synchronous Recurrent Execution Schedule

Algorithmic execution trace across recurrent iterations t = 1, ..., T.

Step	Module / Tensor Operation	Input Representations	Output Representations	Theoretical Purpose
1. Init	Context Projection	AfroXLMR Token Embeddings	Baseline States H(0) and R(0)	Initialize baseline contextual representations
2. ESM (t)	Entity Outer Product	Entity Probability Distribution	ESM Bias Tensor	Generate entity-guided relational constraints
3. Rel Update	Relation Recurrent Cell	Prior Relation State + ESM Bias	Updated Pair Matrix R(t)	Refine token-pair correlation representations
4. RSM (t)	Relational Attention	Pair Matrix R(t) & Entity State	RSM Feedback Vector	Aggregate relational context for each token
5. Ent Update	Entity BiLSTM Layer	Prior Entity State + RSM Feedback	Updated Entity State H_e(t)	Refine token sequence boundary representations
6. Final (T)	CRF & Threshold Decode	Final States H_e(T) and R(T)	Decoded Pairs Set	Jointly emit constrained spans and accepted pairs

Iterating ESM and RSM over T=2 steps allows entity and relation representations to iteratively calibrate each other.

Linear-Chain CRF Score & Multi-Task Objective

Global sequence scoring and supervision timing formulation.

SDRN Linear-Chain CRF & Multi-Task Loss Formulation

$$\text{Score}(X, y) = \sum_{i=1}^{N} E_{i, y_i} + \sum_{i=1}^{N-1} T_{y_i, y_{i+1}} \quad | \quad Z(X) = \sum_{y'} \exp(\text{Score}(X, y'))$$

$$\mathcal{L}_{total} = \left[\log Z(X) - \text{Score}(X, y^*)\right] + \lambda_{rel} \text{BCE}(R^{(T)}, Y^*)$$

- Why CRF is used: Enforces valid sequential transitions (T), eliminating broken tags (e.g., $O \rightarrow I\text{-}ASP$).
- Supervision Timing: Supervised ONLY at final recurrent iteration T; intermediate steps freely synchronize.

Supervising only at step T allows intermediate representations to synchronize freely without premature loss penalties.

Why Everything Is Used & How It Helps

Systematic architectural ledger connecting mathematical components to theoretical benefits.

Component / Layer	Mathematical Formulation	Why It Is Used	How It Helps the Model
AfroXLMR Context	Contextual Sequence Encoder (768-d)	Pretrained Amharic contextualization	Captures long-distance syntactic dependencies
5-Way Entity BiLSTM	Recurrent State with RSM Feedback	Sequential token boundary state	Models left-to-right & right-to-left word context
N × N Relation Matrix	Pair Feature Vector (4d Concat)	Direct token pair modeling	Bypasses word distance; connects separated pairs
ESM Outer Product	Outer Product of Tag Distributions	Injects entity class into relation	Suppresses false relations involving background words
RSM Attention Feedback	Relational Attention Weighted Sum	Injects relation affinity into entity	Rescues ambiguous opinion verbs targeting aspects
Linear-Chain CRF	Global Viterbi Sequence Scoring	Globally normalized sequence decoding	Prevents broken BIO sequences (e.g. O -> I-ASP)
T-Step Supervision	Loss Penalized Exclusively at t = T	Unconstrained intermediate recurrence	Prevents premature convergence on initial noisy states

Every mathematical module in SDRN directly addresses a failure mode of independent cascaded extraction.

Subword Projection & FP16 Stabilization

Engineering solutions implemented to make SDRN train reliably on Amharic text.

Challenge	Engineering Solution	Implementation Details	Impact on Amharic ABSA
Subword Fragmentation	Entity-first subword decoding	decode_subword_predictions_5way()	Rescues prefix B-tags (PR #68)
FP16 NaN Overflow	Attention scaling clamp	Clamp relation logits before exp()	Eliminated training crashes (PR #66)
Scale Mismatch	Uniform decode bio_weights	Scaled Viterbi emissions identically	Matched train & eval objectives (PR #67)
Memory Exhaustion	Gradient checkpointing	AfroXLMR checkpointing enabled	Batch size 8 fit comfortably in VRAM

Subword fragmentation in AfroXLMR was a major root cause of broken spans; entity-first decoding rescued suppressed root tokens.

SDRN Ablation History Across 5 Recorded Runs

Systematic evolution of SDRN configurations evaluated on the held-out test benchmark.

Model Variant	Aspect F1	Opinion Recall	Pair Reachability	Pair F1
AfroXLMR Baseline (Dual BIO)	58.42%	48.32%	35.88%	32.75%
Exact SDRN + 5-Way CRF (Run 1)	60.41%	44.69%	33.04%	34.16%
Asymmetric BIO Weights (Run 2)	59.76%	46.90%	33.51%	33.93%
+ Multi-Class Dice Loss (Run 3)	59.78%	44.66%	33.01%	32.88%
Word-Level SDRN + Pooling	59.96%	44.76%	32.16%	34.07%

CRF Run 1 reached the project peak Aspect F1 (60.41%) and Pair F1 (34.16% @ delta=0.20). Dice loss and asymmetric weights did not surpass it.

AfroXLMR CRF Run 1: Detailed Breakdown

The highest-scoring SDRN model in the repository (tested at optimal threshold delta=0.20).

Evaluation Task	Precision	Recall	F1 Score	True Positives
Aspect Span Extraction	60.60%	60.21%	60.41%	1,766
Opinion Span Extraction	48.67%	44.69%	46.60%	1,318
Joint AOPE Pairs (delta=0.20)	35.88%	32.60%	34.16%	966
Candidate Reachability Ceiling	-	33.04%	-	979 / 2,963 pairs

Aspect F1 reached 60.41%—the highest token-level aspect score across the entire repository. But opinion recall remained low (44.69%).

Asymmetric Class Weighting & Multi-Class Dice Loss

Efforts to alleviate opinion recall suppression through loss modifications.

Run 2: Asymmetric BIO Weights

Hypothesis: Opinions are rarer and more lexically diverse than aspects.

Class weights: Background O at 1.0, Aspects at 1.5, Opinions upweighted to 2.5.

Opinion recall increased slightly to 45.69% (+1.0 percentage point).

Pair F1 dropped to 32.61% (-1.55 points) due to spurious false positive pairs.

Weighting BIO tags shifts precision/recall without expanding underlying span coverage.

Run 3: Multi-Class Dice Loss

Hypothesis: Dice loss optimizes harmonic overlap for extreme foreground imbalance.

Formulation penalizes intersection-over-union directly across token label masks.

Evaluated with auxiliary dice weight 0.5 alongside standard cross-entropy.

Result: Pair F1 dropped to 30.82%, opinion recall dropped to 44.17%.

Dice gradients destabilized token sequence optimization during training.

Loss reweighting shifted the operating point but failed to resolve the core problem: BIO boundary fragmentation.

Subword Mean-Pooling to Eliminate Fragmentation

Moving sequence decisions from tokenizer subwords to authentic morphological words.

Architecture Formulation

AfroXLMR encoder outputs subword hidden states for all constituent pieces.

Constructed a subword-to-word alignment projection matrix.

Applied masked mean pooling across subwords belonging to each lexical word.

All subsequent SDRN layers (BiLSTM, ESM, RSM, CRF) operate on whole words.

Completely eliminates internal subword token boundary artifacts.

Empirical Findings

Aspect F1: 60.20% | Opinion F1: 46.12% | Pair F1: 32.55%.

Results were virtually identical to the subword model (60.41% / 46.60% / 34.16%).

Crucial Discovery: Pooling to whole words did NOT resolve opinion misses.

The remaining error was semantic and lexical, not tokenizer subword splits!

Proved that the token-level BIO paradigm itself was the fundamental bottleneck.

Word-level SDRN proved that errors were not simply tokenizer artifacts; linguistic and phrase-level generalization was the true barrier.

Seen vs. Unseen Vocabulary Diagnostic

Evaluating model recall across exact phrases, token overlaps, and character trigrams.

Vocabulary Granularity	In Train (Seen)	Seen Recall	Unseen in Train	Unseen Recall
Opinion: Exact full phrase	1,014 (34.0%)	59.17% (600)	1,969 (66.0%)	37.63% (741)
Opinion: Token overlap (>= 1)	2,169 (72.7%)	45.78% (993)	814 (27.3%)	42.75% (348)
Opinion: Trigram overlap (>= 50%)	1,450 (48.6%)	54.41% (789)	1,533 (51.4%)	36.01% (552)
Opinion: Multi-word phrase	213 (15.2%)	62.44% (133)	1,184 (84.8%)	31.93% (378)
Aspect: Exact full phrase	1,910 (64.0%)	66.65% (1,273)	1,073 (36.0%)	43.80% (470)
Aspect: Token overlap (>= 1)	2,550 (85.5%)	59.73% (1,523)	433 (14.5%)	50.81% (220)
Aspect: Trigram overlap (>= 50%)	2,326 (78.0%)	63.67% (1,481)	657 (22.0%)	39.88% (262)

72.7% of test opinions share >= 1 token with train, but exact boundary recall collapses on multi-word unseen opinions (31.93%).

The Anatomy of Missed Opinions

Why individual token recognition does not guarantee multi-word span extraction.

The Lexical Recombination Insight

66.0% of test opinions are exact-unseen in training data.

However, only 27.3% (814 spans) are truly novel at the individual token level.

Most unseen opinions are morphological variants or recombinations.

Yet when multi-word opinions are unseen, recall drops to 31.93%.

806 missed unseen multi-word opinions account for 49.1% of all missed opinions.

Why Token BIO Fails on Amharic Phrases

Models easily tag common evaluative tokens (ጥሩ, መጥፎ).

Multi-word civic expressions (e.g. 'ስራ አቁሙ', 'ማስወጣት አለበት') lose continuity.

The tagger predicts B-OPN on the head verb but drops continuation I-OPN.

Token-level BIO cannot evaluate the phrase as an atomic semantic unit.

Unseen multi-word opinions are the single largest source of extraction failure in token-level BIO architectures.

Boundary Misalignment Patterns (`analyze_span_errors.py`)

Categorization of how predicted spans diverge from gold multi-word boundaries.

Error Category	Definition	Prevalence in Multi-Word Opinions	Impact on Pair Extraction
Exact match	Predicted span matches gold boundaries exactly	36.58%	True positive pair candidate
Start matched, cut short	Matches start, but misses continuation tokens	38.42%	Invalidated; pair F1 score = 0
Start matched, overshot	Matches start, but over-extends past gold end	11.20%	Invalidated; spurious boundary
Partial overlap	Overlaps gold tokens but neither boundary matches	8.90%	Zero credit under exact match F1
Completely missed	Zero overlapping tokens tagged as opinion	4.90%	Complete false negative

Over 49% of predicted opinions contain the exact correct start token but fail exact-match evaluation due to boundary clipping.

Pair Coverage Asymmetry (Aspect vs. Opinion)

Measuring conditional failure: what happens to the pair when one span is extracted?

Empirical Pair Breakdown

Total Ground-Truth Pairs: 2,963 explicit pairs in test split.

Both Found (Candidate Ceiling): 979 pairs (33.04%).

Aspect Found, Opinion MISSED: 1,234 pairs (41.65%).

Opinion Found, Aspect MISSED: 312 pairs (10.53%).

Neither Found: 438 pairs (14.78%).

Conditional Attribution

Conditional failure: When the model extracts the Aspect, it misses the Opinion 55.76% of the time!

Attribution of lost pairs: 62.2% of unformed pairs failed solely due to the missing Opinion.

Aspects (nouns/entities) are vastly easier to extract than Opinions (diverse predicates).

The opinion channel is the absolute bottleneck of the system.

Pair formation failure is heavily asymmetric: over 62% of missed pairs failed solely because the opinion was dropped.

The Token-Level Reachability Ceiling

Why SDRN could never achieve high pair recall regardless of relation classification accuracy.

The Two-Stage Metric Hierarchy

1. Candidate Ceiling Recall: Proportion of gold pairs where BOTH spans were extracted.

SDRN Ceiling: 979 / 2,963 = 33.04%.

2. Candidate Conversion Efficiency: Accepted pairs / reachable pairs.

SDRN Conversion: 966 / 979 = 98.67%!

The relation classifier is near-perfect on the spans it actually sees!

The Inevitable Conclusion

SDRN's low pair recall (32.6%) is NOT a relation modeling problem.

The relation classifier accepts 98.67% of reachable pairs.

The ceiling was locked at 33.04% by token BIO boundary errors.

To reach higher pair recall, we must replace token-level BIO tagging entirely.

SDRN achieved 98.67% conversion efficiency on reachable pairs; the entire performance ceiling was dictated by token-level BIO span misses.

Lessons Learned from the SDRN Architecture

What worked, what failed, and the direct engineering decisions it motivated.

Architectural Feature	Empirical Verdict	Why it behaved this way	Next Architectural Step
Joint Recurrent Sync (ESM/RSM)	Effective	Eliminated cascaded 14% pipeline collapse	Retained relational modeling
Linear-Chain 5-Way CRF	Effective	Enforced grammatical tag transitions	Best token-level Aspect F1 (60.41%)
Asymmetric BIO Weights	Ineffective	Traded precision for recall without gain	Abandoned naive weighting
Multi-Class Dice Loss	Ineffective	Destabilized sequence optimization	Returned to cross-entropy
Word-Level Subword Pooling	Diagnostic Only	Proved errors were semantic/phrase-level	Motivated whole-span modeling (Span-ASTE)

SDRN proved joint modeling is essential, but token-level BIO tagging inherently bounds pair recall to ~33%.

The Bridge from SDRN to Span-ASTE

Why token-level tagging was abandoned in favor of span candidate enumeration.

What SDRN Proved

1. Joint extraction solves error compounding (34% vs 14% pipeline F1).
2. CRF decoding yields state-of-the-art aspect extraction (60.41% F1).
3. Relation classification is highly accurate on candidate pairs (98.67% conversion).
4. Unseen multi-word opinions collapse under token-level sequence decoding.

The Shift to Span-ASTE

Problem: Token-level decisions make hard boundary commitments token-by-token.

Solution: Enumerate all contiguous candidate spans up to length 8.

Pool representations over entire spans rather than individual tokens.

Directly test: Can span enumeration bust the 33% candidate ceiling?

Result: Span-ASTE expanded reachability from 33% to 81.91%.

SDRN revealed that boundary errors—not relation classification—were the core bottleneck, directly paving the way for Span-ASTE.

REFERENCES

Bibliography

1. Xu, L., Chia, Y. K., & Bing, L. (2021). Learning Span-Level Interactions for Aspect Sentiment Triplet Extraction. Proceedings of ACL-IJCNLP 2021.

2. Chen, S., Wang, Y., Liu, J., & Wang, Y. (2020). Synchronous Double-Channel Recurrent Network for Aspect-Opinion Pair Extraction. Proceedings of ACL 2020.

3. Cai, H., Xia, R., & Yu, J. (2021). Aspect-Category-Opinion-Sentiment Quadruple Extraction with Implicit Aspects and Opinions. Proceedings of ACL-IJCNLP 2021.

4. Dozat, T., & Manning, C. D. (2017). Deep Biaffine Attention for Neural Dependency Parsing. International Conference on Learning Representations (ICLR).

5. Lin, T.-Y., Goyal, P., Girshick, R., He, K., & Dollár, P. (2017). Focal Loss for Dense Object Detection. Proceedings of ICCV 2017.

6. Xue, L., Barua, A., Constant, N., Al-Rfou, R., Narang, S., Kale, M., Roberts, A., & Raffel, C. (2022). ByT5: Towards a Token-Free Future with Pre-trained Byte-to-Byte Models. Transactions of the ACL (TACL), 10.

7. Raffel, C., Shazeer, N., Roberts, A., Lee, K., Narang, S., Matena, M., Zhou, Y., Li, W., & Liu, P. J. (2020). Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer. Journal of Machine Learning Research (JMLR), 21.

8. Shazeer, N., & Stern, M. (2018). Adafactor: Adaptive Learning Rates with Sublinear Memory Cost. Proceedings of ICML 2018.

9. Conneau, A., Khandelwal, K., Goyal, N., Chaudhary, V., Wenzek, G., Guzmán, F., Grave, E., Ott, M., Zettlemoyer, L., & Stoyanov, V. (2020). Unsupervised Cross-lingual Representation Learning at Scale (XLM-R). Proceedings of ACL 2020.

10. Alabi, J. O., Adelani, D. I., Mosbach, M., & Klakow, D. (2022). Adapting Pre-trained Language Models to African Languages via Multilingual Adaptive Fine-Tuning (AfroXLM-R). Proceedings of COLING 2022.

11. Mekonen, A. (2026). acos-amharic: End-to-End Amharic ACOS Quadruple Extraction [Source code repository]. github.com/EthAIResearch/acos-amharic

12. rasyosef/roberta-base-amharic and Davlan/afro-xlmr-base model cards. Hugging Face Model Hub.